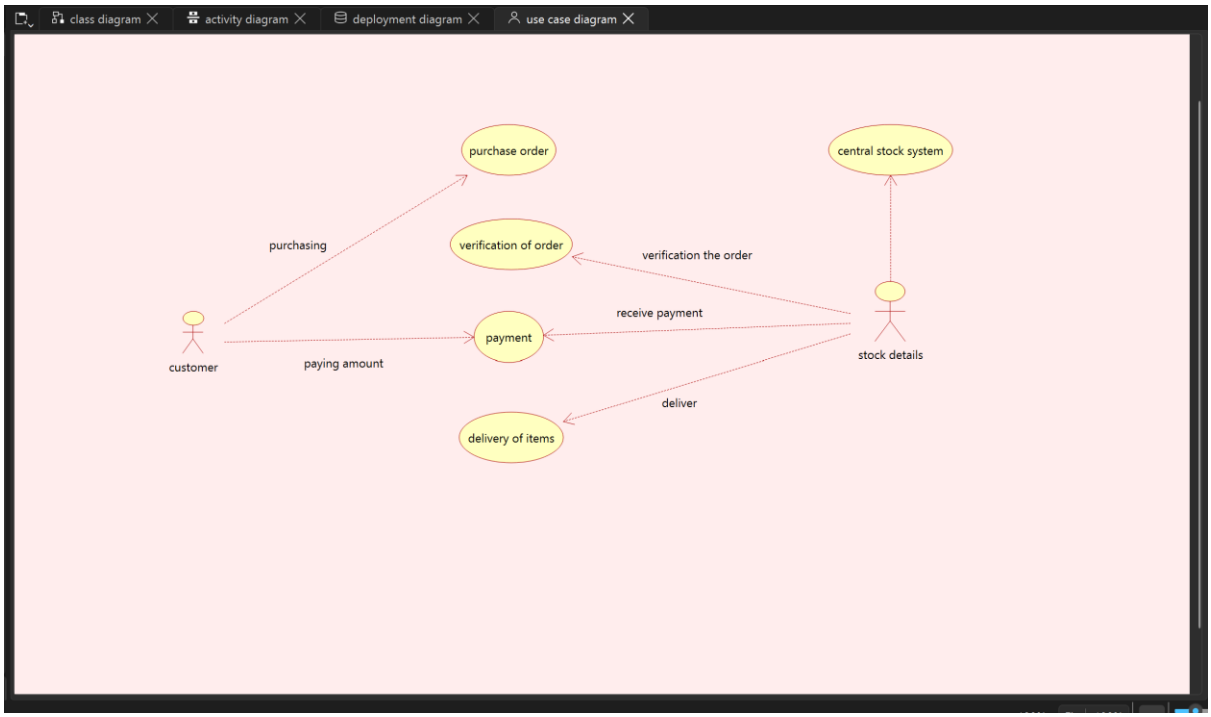
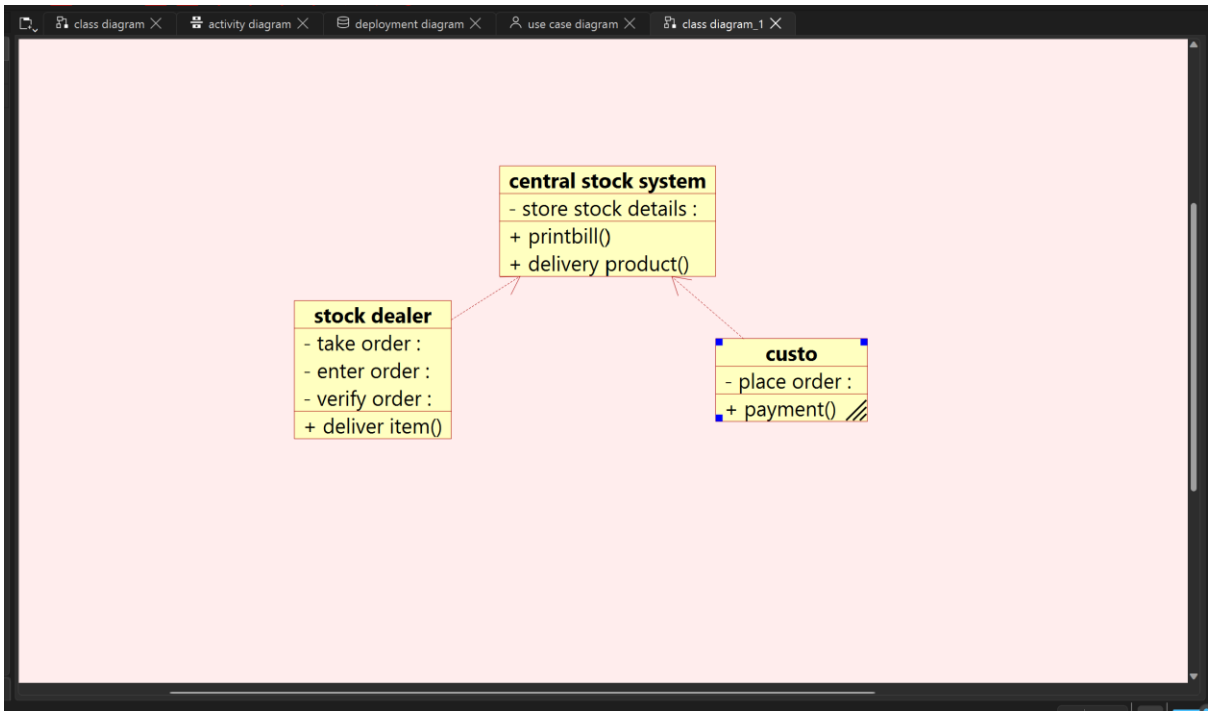
Ex No: 4 – STOCK MAINTAINANCE SYSTEM

NAME: HEMALAKSHMI H (192571049)

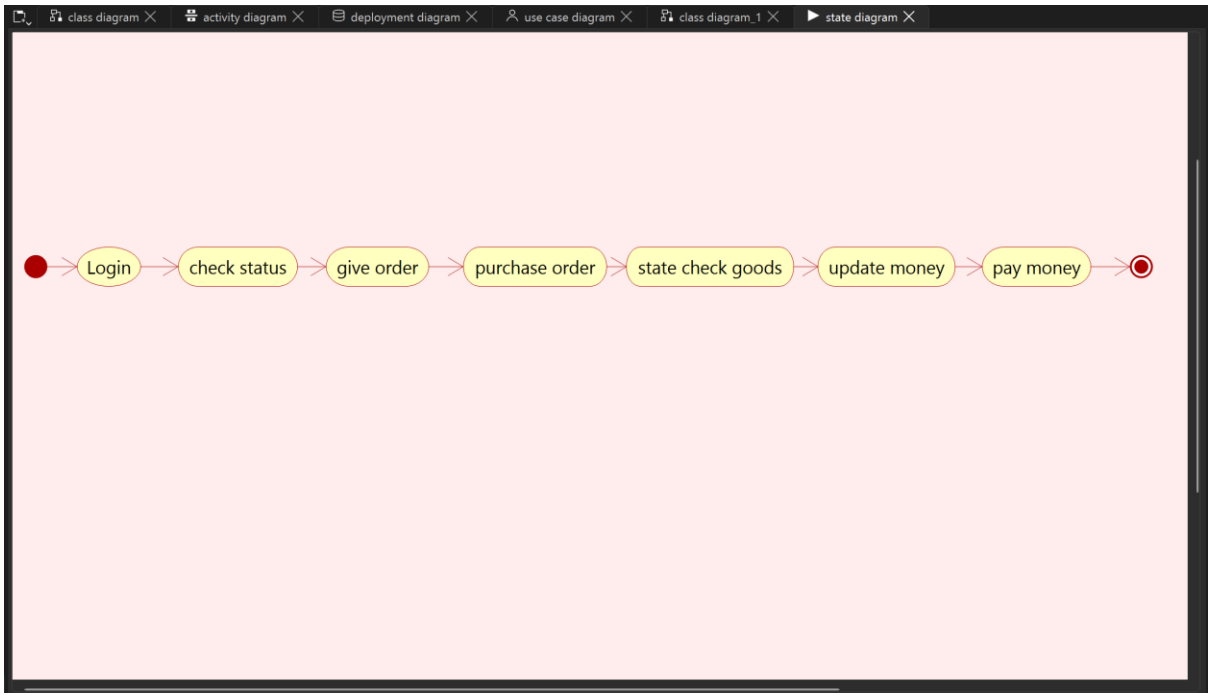
USE CASE DIAGRAM:



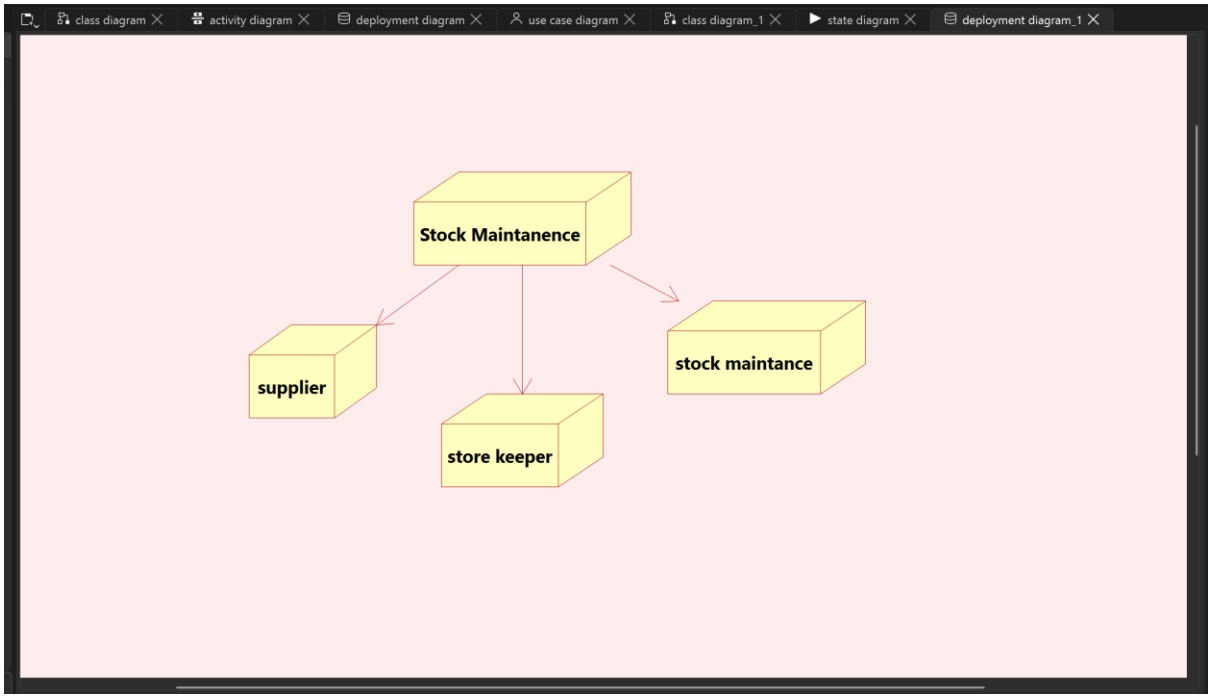
CLASS DIAGRAM:



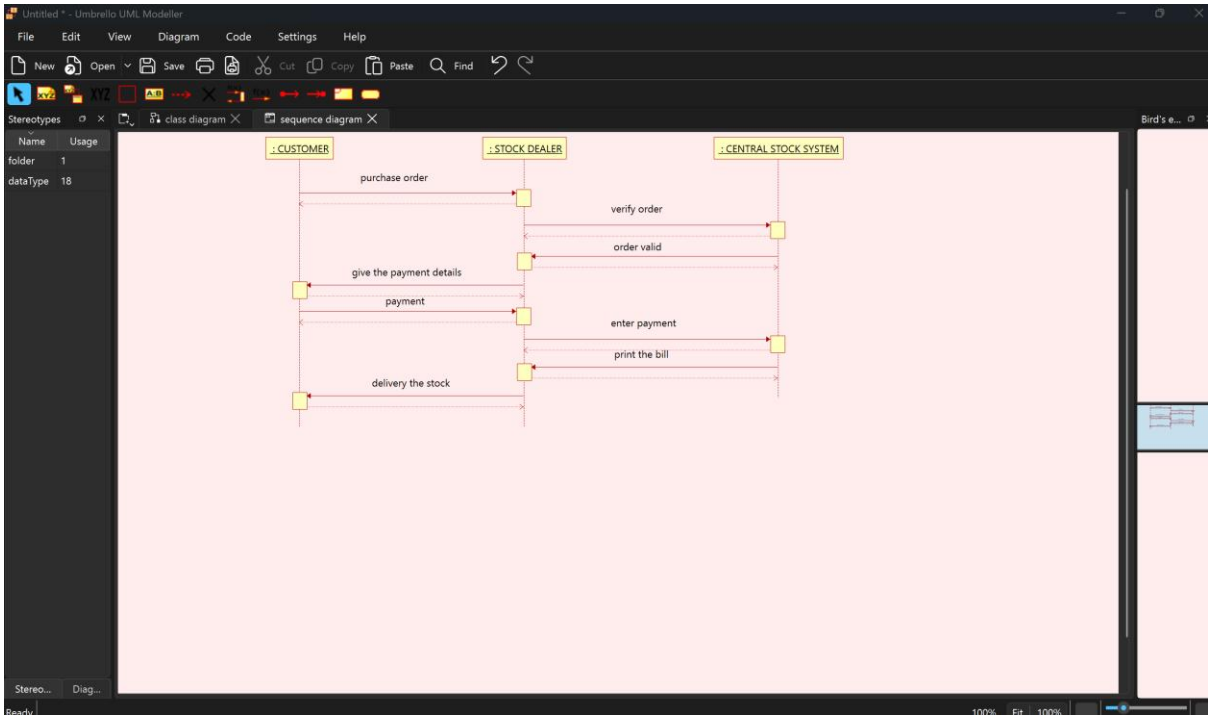
STATE CHART DIAGRAM:



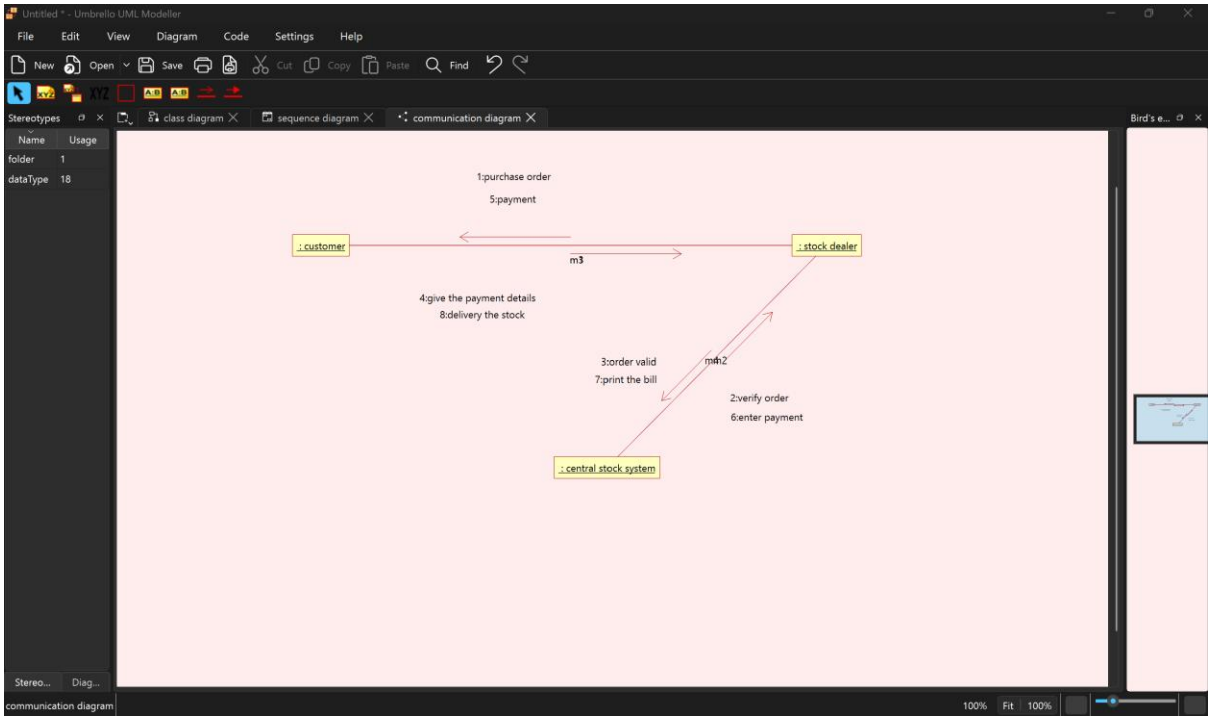
DEPLOYMENT DIAGRAM:



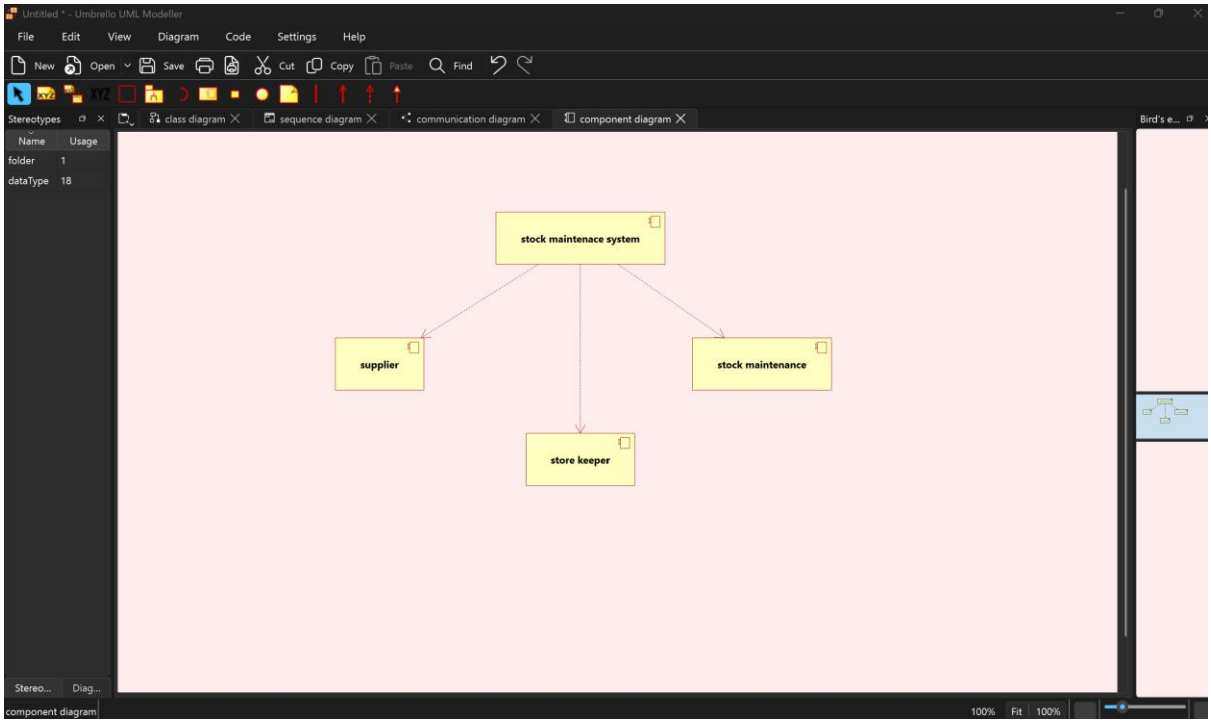
SEQUENCE DIAGRAM:



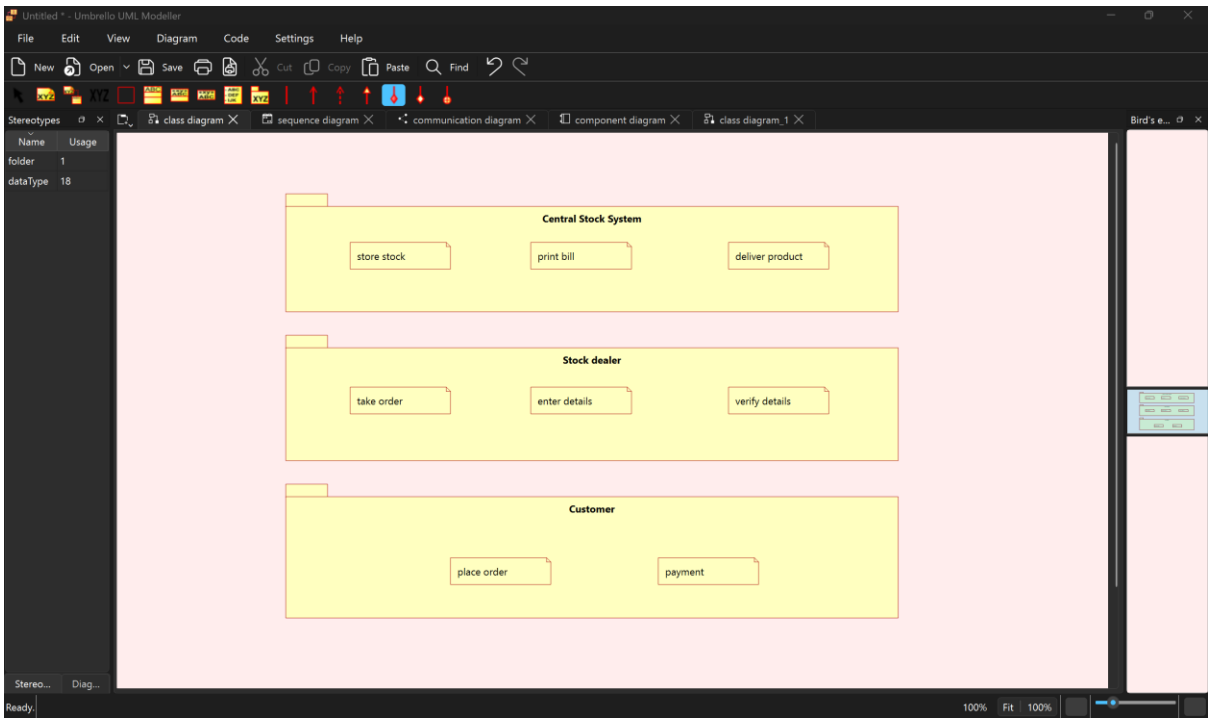
COLLABORATION DIAGRAM:



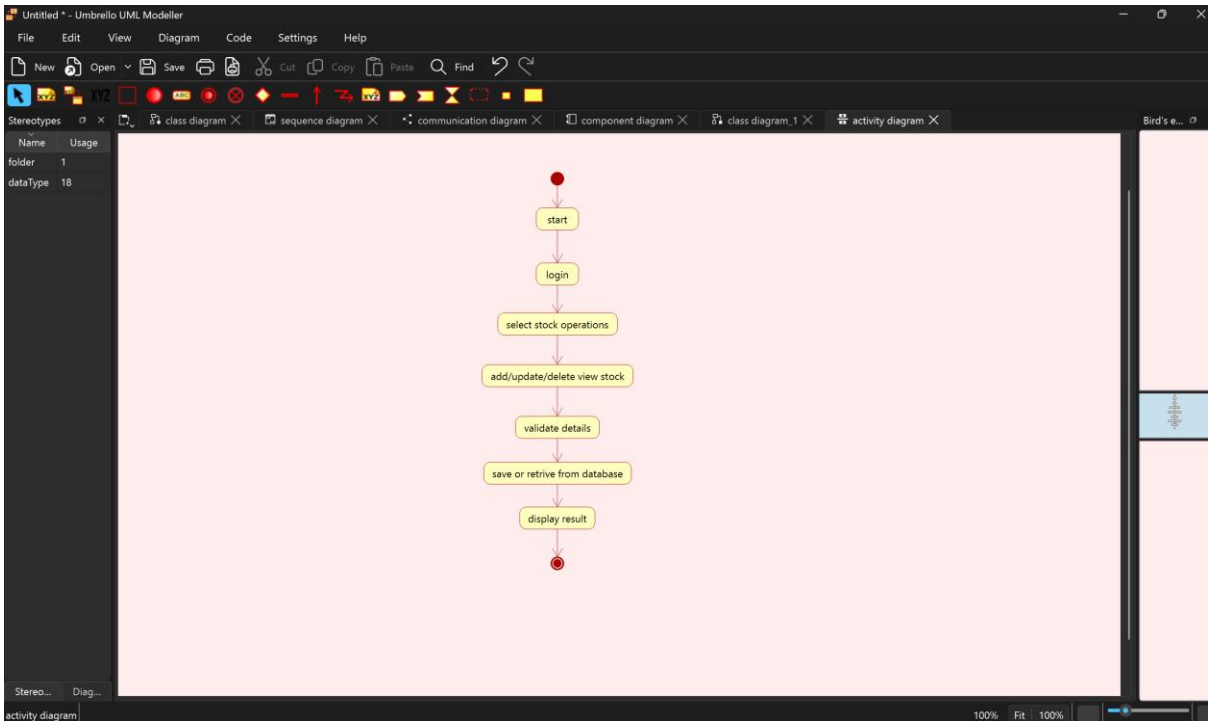
COMPONENT DIAGRAM:



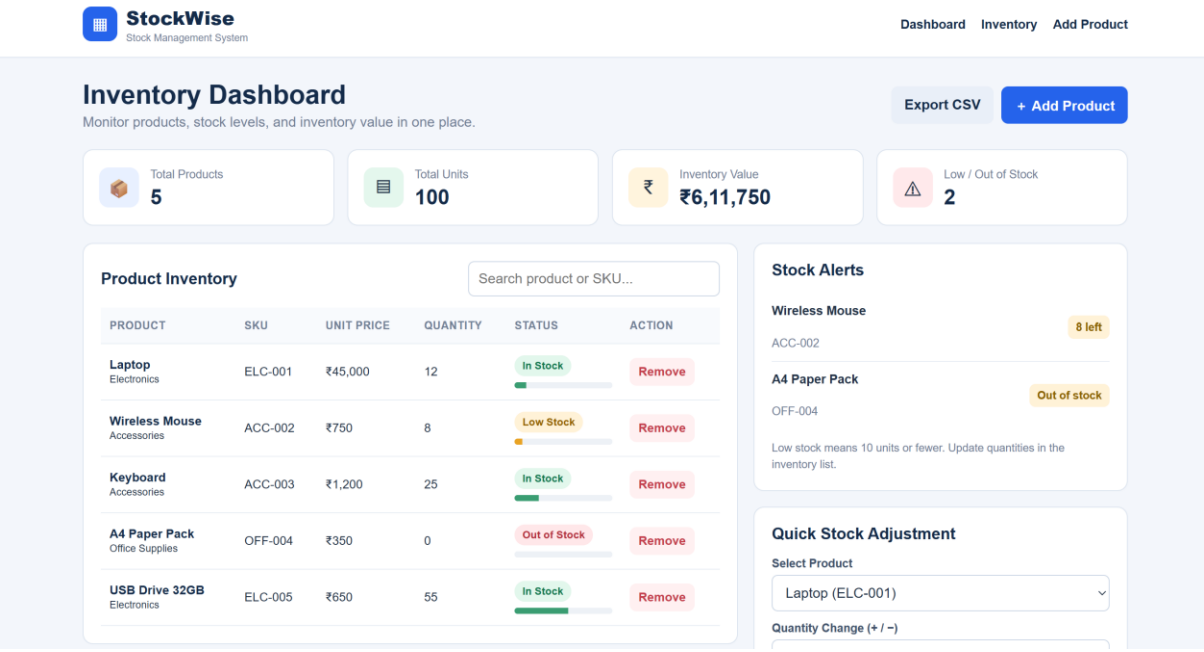
PACKAGE DIAGRAM:

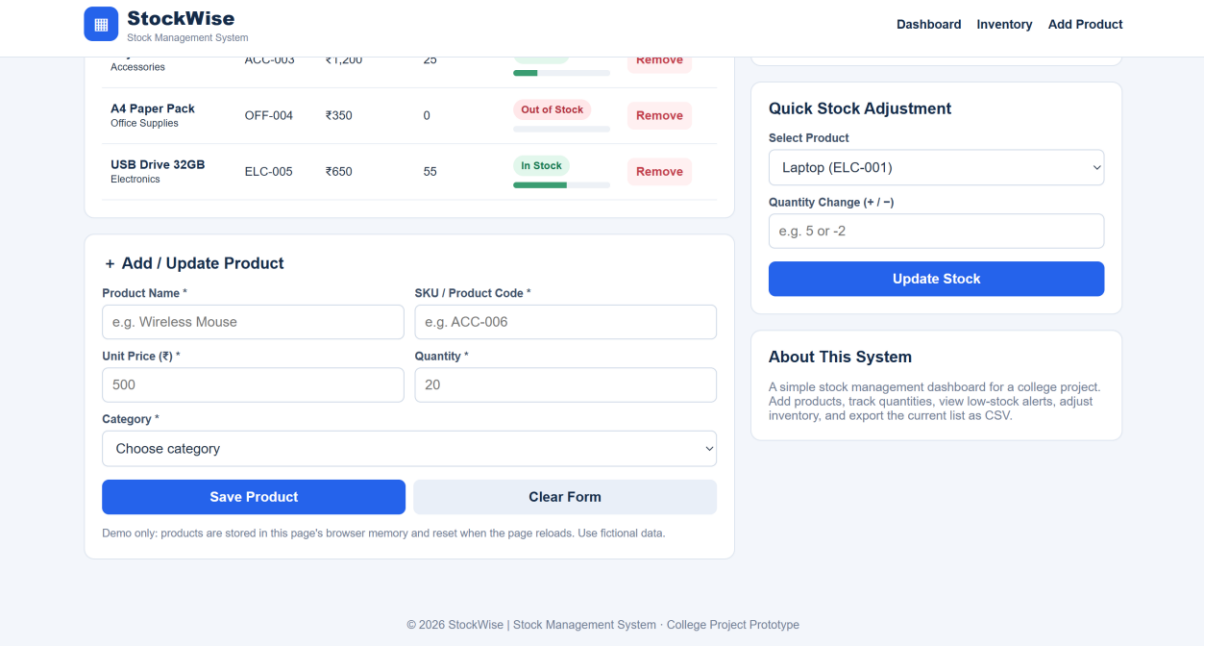


ACTIVITY DIAGRAM:



WEBPAGE FOR STOCK MANAGEMENT SYSTEM





HTML CODE:

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Stock Management System</title>

<style>

*{box-sizing:border-box}

body{margin:0;font-family:Arial,sans-serif;background:#f3f6fb;color:#243047}

header{background:#183b67;color:white;padding:22px}

header h1{margin:0;font-size:25px}

header p{margin:8px 0 0;color:#dce8f7}

main{max-width:1100px;margin:25px auto;padding:0 16px}

.metrics{display:grid;grid-template-columns:repeat(auto-fit,minmax(190px,1fr));gap:15px}

.card,.panel{background:white;border-radius:12px;padding:18px;box-shadow:0 3px 12px #182d4810;margin-bottom:18px}

```
.metric-label{color:#68778c;font-size:14px}
.metric-value{font-size:26px;font-weight:bold;margin-top:10px;color:#183b67}
.layout{display:grid;grid-template-columns:1fr 1fr;gap:18px}
h2{font-size:19px;margin-top:0}
input,select,button{font:inherit;padding:10px;border-radius:7px;border:1px solid #ccd5e2;width:100%;margin:5px 0}
button{background:#245da8;color:white;border:0;cursor:pointer;font-weight:bold}
button:hover{background:#17467f}
button.danger{background:#b83232}
button.secondary{background:#64748b}
.form-grid{display:grid;grid-template-columns:1fr 1fr;gap:8px}
.table-wrap{overflow-x:auto}
table{width:100%;border-collapse:collapse;min-width:650px}
th,td{text-align:left;padding:12px;border-bottom:1px solid #e5eaf1;font-size:14px}
th{background:#edf3fa}
.badge{padding:5px 8px;border-radius:15px;font-size:12px;background:#dcfce7;color:#166534}
.badge.low{background:#fef3c7;color:#92400e}
.badge.out{background:#fee2e2;color:#991b1b}
.alert{padding:10px;border-radius:7px;background:#fff7ed;margin:8px 0;font-size:14px}
.small{font-size:13px;color:#68778c}
@media(max-width:700px){.layout{grid-template-columns:1fr}.form-grid{grid-template-columns:1fr}}
</style>
</head>
<body>
<header>
 <h1>📦 Stock Management System</h1>
 <p>Manage products, monitor stock levels and track inventory value.</p>
</header>

<main>
```

```
<section class="metrics">
 <div class="card">
  <div class="metric-label">Total Products</div>
  <div class="metric-value" id="totalProducts">0</div>
 </div>
 <div class="card">
  <div class="metric-label">Total Units in Stock</div>
  <div class="metric-value" id="totalUnits">0</div>
 </div>
 <div class="card">
  <div class="metric-label">Inventory Value</div>
  <div class="metric-value" id="inventoryValue">₹0</div>
 </div>
 <div class="card">
  <div class="metric-label">Low / Out of Stock</div>
  <div class="metric-value" id="stockAlerts">0</div>
 </div>
</section>

<section class="layout">
 <div class="panel">
  <h2>Add New Product</h2>
  <form id="productForm">
   <input id="name" placeholder="Product name" required>
   <div class="form-grid">
    <input id="sku" placeholder="SKU / Product code" required>
    <select id="category" required>
     <option value="">Select category</option>
     <option>Electronics</option>
     <option>Stationery</option>
     <option>Food</option>
```

```
        <option>Clothing</option>
        <option>Other</option>
      </select>
      <input id="price" type="number" min="0.01" step="0.01"
          placeholder="Unit price (₹)" required>
      <input id="quantity" type="number" min="0" step="1"
          placeholder="Quantity" required>
    </div>
    <button type="submit">Add Product</button>
   </form>
   <p class="small">Enter a unique SKU for each product.</p>
  </div>

  <div class="panel">
   <h2>Stock Adjustment</h2>
   <form id="adjustForm">
    <select id="adjustProduct" required>
     <option value="">Choose a product</option>
    </select>
    <input id="adjustQty" type="number" step="1"
        placeholder="Quantity change (e.g. 5 or -2)" required>
    <button type="submit">Update Stock</button>
   </form>
   <p class="small">Use a positive number to add stock and a negative number to
remove stock.</p>
   <h2 style="margin-top:22px">Stock Alerts</h2>
   <div id="alerts"><p class="small">No alerts.</p></div>
  </div>
 </section>

 <section class="panel">
```

```
  <h2>Product Inventory</h2>
  <input id="search" placeholder="Search by product name, SKU or category">
  <div class="table-wrap">
   <table>
    <thead>
     <tr>
      <th>Product</th>
      <th>SKU</th>
      <th>Category</th>
      <th>Unit Price</th>
      <th>Quantity</th>
      <th>Total Value</th>
      <th>Status</th>
      <th>Action</th>
     </tr>
    </thead>
    <tbody id="productTable"></tbody>
   </table>
  </div>
  <button id="exportBtn" style="max-width:200px;margin-top:15px">
   Export to CSV
  </button>
 </section>
 <p class="small">College project demo • Data is stored temporarily in the browser and resets when the page reloads.</p>
</main>

<script>
let products = [
 {name:"Wireless Mouse",sku:"ELEC001",category:"Electronics",price:599,quantity:15},
 {name:"Notebook",sku:"STAT001",category:"Stationery",price:50,quantity:8},
```

```
 {name:"USB Cable",sku:"ELEC002",category:"Electronics",price:199,quantity:0},
 {name:"Pen Pack",sku:"STAT002",category:"Stationery",price:120,quantity:25}
];

const $ = id => document.getElementById(id);
const money = amount => "₹" + Number(amount).toLocaleString("en-IN", {
 minimumFractionDigits:2, maximumFractionDigits:2
});

function statusOf(product){
 if(product.quantity === 0) return "Out of Stock";
 if(product.quantity <= 10) return "Low Stock";
 return "In Stock";
}

function render(){
 $("totalProducts").textContent = products.length;
 $("totalUnits").textContent =
  products.reduce((sum,p)=>sum+p.quantity,0);
 $("inventoryValue").textContent = money(
  products.reduce((sum,p)=>sum+p.price*p.quantity,0)
 );
 $("stockAlerts").textContent =
  products.filter(p=>p.quantity<=10).length;

 const search = $("search").value.toLowerCase();
 const filtered = products.filter(p =>
  [p.name,p.sku,p.category].some(v=>v.toLowerCase().includes(search))
 );

 $("productTable").innerHTML = filtered.map(p => {
```

```
  const status = statusOf(p);
  const badge = p.quantity===0 ? "out" : p.quantity<=10 ? "low" : "";
  return `<tr>
   <td>${escapeHTML(p.name)}</td>
   <td>${escapeHTML(p.sku)}</td>
   <td>${escapeHTML(p.category)}</td>
   <td>${money(p.price)}</td>
   <td>${p.quantity}</td>
   <td>${money(p.price*p.quantity)}</td>
   <td><span class="badge ${badge}">${status}</span></td>
   <td><button class="danger"
onclick="removeProduct('${p.sku}')">Remove</button></td>
  </tr>`;
 }).join("");

 if(filtered.length===0){
  $("productTable").innerHTML =
   '<tr><td colspan="8">No products found.</td></tr>';
 }

 $("adjustProduct").innerHTML =
  '<option value="">Choose a product</option>' +
  products.map(p=>`<option value="${escapeHTML(p.sku)}">${escapeHTML(p.name)}
(${escapeHTML(p.sku)})</option>`).join("");

 const alerts = products.filter(p=>p.quantity<=10);
 $("alerts").innerHTML = alerts.length ? alerts.map(p =>
  `<div class="alert"><strong>${escapeHTML(p.name)}</strong>: ${
   p.quantity===0 ? "Out of stock" : p.quantity+" units remaining"
  }</div>`
 ).join("") : '<p class="small">All products have sufficient stock.</p>';
```

```
}

function escapeHTML(value){
 return String(value).replace(/[&<>"']/g, c => ({
  "&":"&amp;","<":"&lt;",">":"&gt;","'":"&quot;","'":"&#39;"
 }[c]));
}

$("productForm").addEventListener("submit", function(e){
 e.preventDefault();
 const name = $("name").value.trim();
 const sku = $("sku").value.trim();
 const category = $("category").value;
 const price = Number($("price").value);
 const quantity = Number($("quantity").value);

 if(products.some(p=>p.sku.toLowerCase()===sku.toLowerCase())){
  alert("This SKU already exists. Please use a unique SKU.");
  return;
 }
 if(!name || !sku || !category || price<=0 ||
   !Number.isFinite(price) || !Number.isInteger(quantity) || quantity<0){
  alert("Please enter valid product details.");
  return;
 }

 products.push({name,sku,category,price,quantity});
 this.reset();
 render();
 alert("Product added successfully!");
});
```

```
$("adjustForm").addEventListener("submit", function(e){
 e.preventDefault();
 const sku = $("adjustProduct").value;
 const change = Number($("adjustQty").value);
 const product = products.find(p=>p.sku===sku);

 if(!product || !Number.isInteger(change) || change===0){
  alert("Choose a product and enter a non-zero whole number.");
  return;
 }
 if(product.quantity+change<0){
  alert("Stock quantity cannot become negative.");
  return;
 }

 product.quantity += change;
 $("adjustQty").value = "";
 render();
 alert("Stock updated successfully!");
});

function removeProduct(sku){
 const product = products.find(p=>p.sku===sku);
 if(product && confirm("Remove "+product.name+" from inventory?")){
  products = products.filter(p=>p.sku!==sku);
  render();
 }
}

$("search").addEventListener("input", render);
```

```
$("exportBtn").addEventListener("click", function(){
 const headers = ["Name","SKU","Category","Unit Price","Quantity","Total Value","Status"];
 const rows = products.map(p=>[
  p.name,p.sku,p.category,p.price,p.quantity,
  (p.price*p.quantity).toFixed(2),statusOf(p)
 ]);
 const csv = [headers,...rows].map(row =>
  row.map(v=>'"'+String(v).replace(/"/g,'""')+'"').join(",")
 ).join("\r\n");

 const blob = new Blob(["\uFEFF"+csv],{type:"text/csv;charset=utf-8;"});
 const url = URL.createObjectURL(blob);
 const link = document.createElement("a");
 link.href = url;
 link.download = "stock_inventory.csv";
 link.click();
 URL.revokeObjectURL(url);
});

render();
</script>
</body>
</html>
```